INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, Third Semester Examination, November - 2024

Subject Code: BCACC302

Subject Name: Object Oriented Programming with C++

Full Marks: 30 Duration: 1 Hour

Part - A

Attempt 5 questions

Each question carries 2 marks (2 × 5)

1. Differentiate between default constructor and a parameterized constructor. 2

or

What is a copy constructor? How is it different from a regular constructor? 2

2. Define polymorphism. 2

or

Define virtual function. 2

3. Define hierarchical inheritance. 2

or

Define hybrid inheritance. 2

4. Explain the advantages of using exception handling mechanism in a program. 2

or

Explain different types of exceptions. 2

5. Define base class and derived class in C++. 2

or

Define virtual base class and its uses. 2

Part - B

Attempt 2 questions

Each question carries 5 marks (5 × 2)

6. Write a class Person with multiple constructors to handle different initialization scenarios. 5

or

Write a program that demonstrates exception handling using try, catch, and throw blocks. 5

7. Differentiate between compile time polymorphism and run time polymorphism. 5

or

Explain different types of Inheritance with help of sample example. **5**

Part - C

Attempt 1 question

Each question carries 10 marks (10 × 1)

8. Write down a C++ program to swap two numbers without using third variable with the help of friend function. **10**

or

Write a C++ program to demonstrate the concept of single inheritance. Create a base class Person with common attributes like name and age, and a derived class Student that adds specific attributes like roll number and marks. Implement functions to display the details of a student. **10**

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